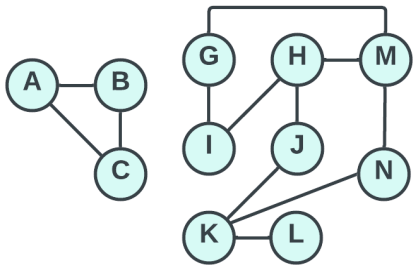
Assignment 1
Due at 11:59 pm on Wednesday, September 16

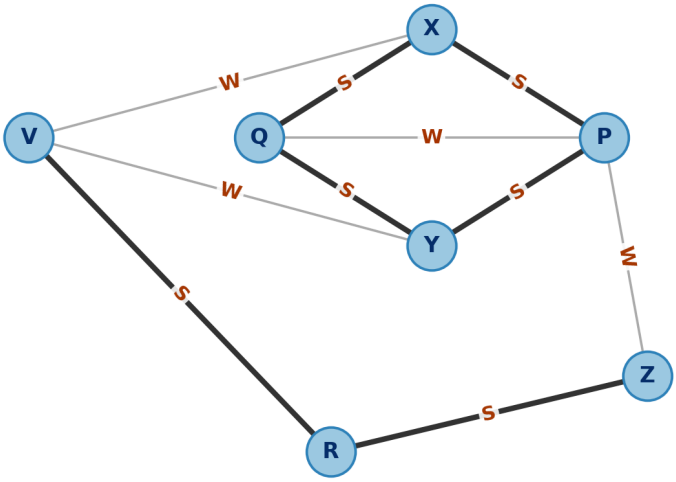
Reading: Chapters 1, 2, and 3 of the textbook

1. (21 points) Consider the shown network:



   a. Identify the longest cycles in the network.
   b. Is the network connected? Explain.
   c. How many connected components does the network have? Which nodes are in each component?
   d. How many nodes are in the largest component of the network?
   e. Find the distance from node H to every other node in the network using breadth-first search. Draw a diagram of the breadth-first search procedure similar to the one shown in class, where each new layer of nodes is identified and circled. *Remember, the BFS diagram should have all of the nodes in it!*
   f. What is the longest cycle you can make by adding one edge to the network? Please specify what edge you would add, how long the longest resulting cycle is, and the cycle itself (i.e., the sequence of nodes). You do not have to explain how you found the edge and the cycle.

2. (22 points) In the network below, the ties (P, X), (P, Y), (Q, X), (Q, Y), (R, V), and (R, Z) are labeled strong (S), while all others are labeled weak (W).



a. Identify all nodes that violate the Strong Triadic Closure Property. For each one, name its strong ties and the missing edge.
b. A student says that fixing this network requires the same number of new edges as the number of violations. Is the student right? Explain. Find a smallest set of edges such that adding them makes the network satisfy the Strong Triadic Closure Property. Explain why no smaller set of edges can work.
c. List all local bridges in the original network together with their spans. Then list all local bridges again after adding the edges from part (b). Explain why the list changed.
d. Compute the average local clustering coefficient of the network before and after adding the edges identified in part (b). Does the direction of the change match what triadic closure predicts? Explain.

3. This problem has 2 parts

Part 1 (10 points, From Textbook Section 3.7 - Problem 2): Consider the graph in Figure 3.21, in which each edge — except the edge connecting b and c — is labeled as a strong tie (S) or a weak tie (W). According to the theory of strong and weak ties, with the strong triadic closure assumption, how would you expect the edge connecting b and c to be labeled? Give a brief (1-3 sentence) explanation for your answer.

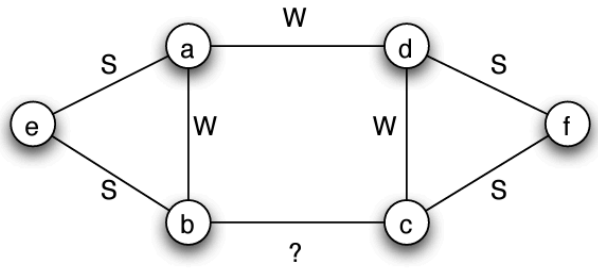


Figure 3.21:

Part 2 (10 points, From Textbook Section 3.7 - Problem 3): In the social network depicted in Figure 3.22, with each edge labeled as either a strong or weak tie, which nodes satisfy the Strong Triadic Closure Property from Chapter 3, and which do not? Provide an explanation for your answer.

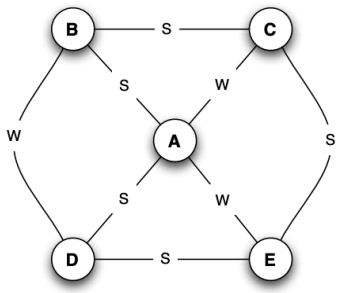


Figure 3.22:

4. (20 points) Consider an undirected graph with 10 nodes. If the graph has exactly one local bridge (A,B), is it possible for the graph to also satisfy each of the following properties? If it's possible, draw a network and explain how it satisfies the properties. If it is not possible, explain why it is impossible to satisfy all properties.
    a. Edge (A,B) is also a bridge
    b. Edge (A,B) has span 4
    c. Nodes A and B both have a local clustering coefficient greater than 0
    d. Edge (A,B) has an embeddedness of 2

    Note: You aren't trying to satisfy properties a-d at the same time, but satisfy each, one at a time – e.g., if you think part (a) is possible, you'll need to draw a simple, undirected graph with 10 nodes that has one edge labeled (A,B) that is a local bridge and also a bridge and no other edges that are local bridges; and you need to show that the network satisfies all of these conditions.

5. (17 points) Consider an undirected, connected graph with n>2 nodes. If a node has a local clustering coefficient (LCC) of 0, prove that every edge connected to that node is a local bridge.
    Hint: Think about the definitions: a local bridge is an edge whose removal increases the shortest distance between its endpoints from 1 to at least 3, and a node's LCC = 0 means none of its neighbors are connected. Try proving the statement by contradiction using these ideas.